Devansh Mistry

AI Application Engineer

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Portfolio | GitHub | LinkedIn

Professional Summary

AI Application Engineer and Data Scientist focused on building performant and reliable LLM applications that drive business efficiency. Specialized in designing complex Agentic AI flows, integrating Voice Agents, and deploying scalable Retrieval-Augmented Generation (RAG) pipelines into modern web environments. Experienced in developing Proof of Concepts (POCs), evaluating AI agent outputs for strict alignment, and implementing cloud-native MLOps architectures to transition rapid prototypes into production-grade ML systems.

Technical Skills

Generative AI	Large Language Models (LLMs), Agentic AI, RAG Pipelines, Prompt/Context Engineering, Parameter-Efficient Fine-Tuning (PEFT), Open-source LLM deployment (LLaMA, Ollama), NLP, Semantic Search.
Cloud & MLOps	Microsoft Azure AI Foundry, Google Cloud Platform (GCP), Vertex AI, Model Deployment & Monitoring, CI/CD for ML, Cost Optimization, Docker, GitHub Actions.
Data & DBs	Python, SQL, Vector Databases (MongoDB Atlas, Chroma), Data Pipelines, ETL Processes, Knowledge Retrieval Systems.
Software Eng.	React.js, Next.js, REST APIs, Microservices Architecture, QA/QE Automation Testing, Google AI Agent SDK.
Microsoft	Microsoft 365 Copilot capabilities, Azure OpenAI.

Featured Projects

2026 **Tenant-Defender (Zero-Hallucination Legal AI)**, *TypeScript, Next.js, FastAPI, Gemini API, Strict-CAG*, [GitHub]
- Built a zero-hallucination legal AI system (GenAI Genesis 2026) using Strict Cache-Augmented Generation (CAG) and Agentic OCR to analyze Ontario tenancy documents.
- Engineered automated document parsing and deterministic statutory validation pipelines to protect renters from bad-faith eviction notices with strict legal grounding.

2025 **Real-Time AI Voice Agent**, *Python, Gemini AI, ElevenLabs, Twilio/WebSocket*, [GitHub]
- Architected a dual-tier voice AI orchestrator featuring a high-fidelity ElevenLabs tier (<150ms latency) and a cost-optimized EdgeTTS/FastAPI WebSocket pipeline (<500ms latency).
- Engineered dynamic intent routing across deterministic Tool Calling (live booking), local RAG (ChromaDB policy retrieval), and real-time interruption handling (VAD).
- Built an asynchronous event-driven CRM logging pipeline that syncs structured call summaries to Google Sheets without blocking the main execution thread.

2025 **Smart Recipe Retrieval (RAG System)**, *Python, VectorDB, Gemini API*, [GitHub]
- Built a comprehensive RAG pipeline using MongoDB Atlas Vector Search and Google Gemini LLM to generate highly tailored recommendations.
- Implemented semantic retrieval and prompt engineering optimization to dramatically improve contextual accuracy in the final LLM output.

2025 **Agent-scope (AI Observability Platform)**, *TypeScript, React, MLOps*, [GitHub]
- Developed a client-side agent observability platform to monitor LLM application performance, track agent routing decisions, and visualize real-time execution flow.
- Established foundational MLOps tracking capabilities to analyze cost optimization and risk mitigation during complex agent workflows.

2026 **Autonomous B2B Procurement Agent**, *Google AI Agent SDK, Next.js, Local LLMs*, [GitHub]
- Developed an autonomous Agentic AI system for B2B procurement and negotiation, designing multi-step agent flows.
- Implemented robust tool-use capabilities and integrated open-source LLMs to mitigate data privacy risks.
- Built backend tools and abstractions to manage complex negotiation logic within a scalable Next.js microservices environment.

Professional Experience

Apr 2026–Aug 2026 **AI/ML Evaluator Specialist (Remote)**, *Mercor*

- Evaluated and ranked AI-generated outputs against complex user prompts to ensure strict alignment with intended style, tone, and information needs.
- Authored detailed explanatory statements and justifications for ranking decisions, providing high-quality human preference data used for model refinement.
- Conducted manual data annotation and quality assurance for AI responses, identifying errors and ensuring adherence to project guidelines.

May 2022–Feb 2023 **Data Scientist Intern (Remote)**, *iNeuron.ai*

- Deployed predictive machine learning models on Azure ML, reducing manual analysis time for internal teams by automating data insights.
- Optimized data preprocessing pipelines, conducted systematic hyperparameter tuning, reducing feature engineering latency by 20% and improving model inference accuracy.
- Containerized models using Flask + Docker, automating deployment and scaling across environments.
- Designed interactive dashboards to visualize performance metrics and streamline insights for decision-makers.

Jan 2023–Feb 2023 **Machine Learning Intern (Full-time)**, $F(x)$ *Data Labs*, Ahmedabad, India

- Prototyped NLP and predictive models, developing rapid Proof of Concepts (POCs) to validate feasibility for real-world aviation operations.
- Collaborated intensely with cross-functional stakeholders, translating complex ML outputs into actionable business strategies to support senior leadership in AI adoption.

Education

2024–2025 **Post-Graduate Certificate, Web Development**, *Conestoga College*, Waterloo, ON

2019–2023 **Bachelor of Engineering, Information Technology**, *L.D. College of Engineering*, India
Relevant Coursework: Data Science & Artificial Intelligence, Pattern Recognition, Probability & Statistics

Certifications & Training

Active Microsoft Certified: Azure AI Engineer Associate (AI-102)

Active Microsoft Certified: Azure Data Scientist Associate (DP-100)

Active Microsoft Certified: Azure AI Fundamentals (AI-900)

Active IBM Deep Learning and Reinforcement Learning

Active IBM Machine Learning Specialization

Study Deep Learning & Neural Networks (Andrej Karpathy Curriculum)

Pursuing Google Cloud Professional Machine Learning Engineer